

Figure 1: Part (a): right rectangular loop.

The force can also be explicitly calculated for this configuration. See scanned solution for detail.

(c) Let us first set up the configuration from scratch. Consider a ring originally in the xz -plane, with center at the origin. Points on this ring can be parameterized by angle θ . Then, for a point on the ring, we have its location and the corresponding current density at this point as

$$\mathbf{r} = (a \cos \theta, 0, a \sin \theta), \quad \mathbf{I}_1 = -I_1 \hat{\theta} = I_1(\sin \theta, 0, -\cos \theta).$$

Now, rotate this ring around the z -axis by an angle of $\pi/2 - \alpha$, then

$$\mathbf{r} = (a \cos \theta \sin \alpha, -a \cos \theta \cos \alpha, a \sin \theta), \quad \mathbf{I}_1 = I_1(\sin \theta \cos \alpha, -\sin \theta \sin \alpha, -\cos \theta).$$

Finally, move the center of the ring along y -axis by a distance of d , and

$$\mathbf{r} = (a \cos \theta \sin \alpha, d - a \cos \theta \cos \alpha, a \sin \theta),$$

while $\mathbf{I}_1$ stays the same. In fact, our experience in part (b) suggests that we should also move the ring by an ϵ of distance in x -direction. Therefore, points on the ring will be

$$\mathbf{r} = (\epsilon + a \cos \theta \sin \alpha, d - a \cos \theta \cos \alpha, a \sin \theta),$$

and we will set ϵ to 0 in the final step.

From part (a), the interaction energy is

$$W_{12} = \oint \mathbf{J}_1 \cdot \mathbf{A} dl = \oint (-\hat{z} I_1 \cos \theta) \cdot \left(\hat{z} \frac{\mu_0 I_2}{2\pi} \log r \right) dl = -\frac{\mu_0 I_1 I_2 a}{2\pi} \int_0^{2\pi} \cos \theta \log r d\theta,$$

where r is the distance from a point on the ring to the z -axis,

$$r^2 = (\epsilon + a \cos \theta \sin \alpha)^2 + (d - a \cos \theta \cos \alpha)^2 = |d + i\epsilon - a \cos \theta e^{-i\alpha}|^2,$$

and

$$\log r = \log |d + i\epsilon - a \cos \theta e^{-i\alpha}| = \text{Re} [\log (d + i\epsilon - a \cos \theta e^{-i\alpha})].$$

This is the key to the final desired form of solution to this problem.

Now,

$$W_{12} = -\frac{\mu_0 I_1 I_2 a}{2\pi} \text{Re} \left[\int_0^{2\pi} \cos \theta \log (d + i\epsilon - a \cos \theta e^{-i\alpha}) d\theta \right],$$

and we need to perform the integral,

$$\begin{aligned} Q &= \int_0^{2\pi} \cos \theta \log (p - q \cos \theta) d\theta \\ &= \sin \theta \log (p - q \cos \theta) \Big|_0^{2\pi} - \int_0^{2\pi} \frac{q \sin^2 \theta}{p - q \cos \theta} d\theta \\ &= -\frac{2\pi p}{q} \left(1 - \sqrt{1 - \frac{q^2}{p^2}} \right), \end{aligned}$$

where we have used Gradshteyn and Ryzhik, 8th ed., 3.644.1, pg. 404 (not found), with $|q| < |p|$.

Therefore, with nonzero ϵ , $p = d + i\epsilon$ and $q = ae^{-i\alpha}$,

$$W_{12} = \mu_0 I_1 I_2 \cdot \text{Re} \left\{ (d + i\epsilon) \left[e^{i\alpha} - \sqrt{e^{2i\alpha} - \frac{a^2}{(d + i\epsilon)^2}} \right] \right\}.$$

Setting ϵ to 0, we will obtain the desired result.

The force can be calculated in the same way as in part (b),

$$\begin{aligned}\mathbf{F} &= \left(\frac{\partial W_{12}}{\partial \epsilon}, \frac{\partial W_{12}}{\partial d} \right) \Big|_{\epsilon=0} \\ &= \mu_0 I_1 I_2 \begin{pmatrix} -\text{Im} \left[e^{i\alpha} - \frac{e^{2i\alpha}}{\sqrt{e^{2i\alpha} - a^2/d^2}} \right] \\ \text{Re} \left[e^{i\alpha} - \frac{e^{2i\alpha}}{\sqrt{e^{2i\alpha} - a^2/d^2}} \right] \end{pmatrix}^\top.\end{aligned}$$

(d) In the limit $d \gg a, b$, we have

$$\sqrt{e^{2i\alpha} - \frac{a^2}{d^2}} \approx e^{i\alpha} - \frac{a^2}{2d^2} e^{-i\alpha},$$

and

$$\log \left(d^2 \pm bd \cos \alpha + \frac{b^2}{4} \right) \approx \log d^2 \pm \frac{b}{d} \cos \alpha.$$

Then, for the circular loop,

$$W_{12} = \mu_0 I_1 I_2 d \cdot \text{Re} \left[\frac{a^2}{2d^2} e^{-i\alpha} \right] = \frac{\mu_0 I_2}{2\pi d} \cdot \pi a^2 I_1 \cdot \cos \alpha = \mathbf{m} \cdot \mathbf{B},$$

and for the rectangular loop,

$$W_{12} = \frac{\mu_0 I_1 I_2 a}{4\pi} \cdot \frac{2b}{d} \cos \alpha = \frac{\mu_0 I_2}{2\pi d} \cdot ab I_1 \cdot \cos \alpha = \mathbf{m} \cdot \mathbf{B}.$$